Unit: Unit 3: Functions
Topic: Operations with Functions

Name: ____________
Date: __________

Homework (Jalapeno)

Q1. If $f(x) = x + 2$ and $g(x) = 3x - 1$, what is $(f + g)(x)$?
(A) $4x + 1$
(B) $4x - 1$
(C) $3x + 1$
(D) $3x - 3$
(E) $2x + 1$

Q2. What is the domain of $f(x) = \frac{1}{x}$ and $g(x) = \frac{1}{x-2}$ when combined as $f(g(x))$?
(A) $(-\infty, -2) \cup (-2, \infty)$
(B) $(-\infty, 2) \cup (2, \infty)$
(C) $(-\infty, 0) \cup (0, \infty)$
(D) $(-\infty, 0) \cup (2, \infty)$
(E) $(-2, 2)$

Q3. The ancient Egyptians used the function $f(x) = 2x + 1$ to calculate the area of a triangle. If they used another function $g(x) = x - 3$, what is $(f - g)(x)$?
(A) $x + 4$
(B) $x - 2$
(C) $3x - 2$
(D) $x + 2$
(E) $2x - 2$

Q4. The function $f(x) = x^2 + 1$ represents the number of workers needed to build a pyramid. If $g(x) = 2x - 1$, what is $(f \cdot g)(x)$?
(A) $2x^3 - x^2 + 2x - 1$
(B) $2x^3 + x^2 - 2x - 1$
(C) $2x^3 - x^2 - 2x + 1$
(D) $2x^3 + x^2 + 2x + 1$
(E) $x^3 - 2x^2 + x - 1$

Q5. The ancient Greeks used the function $f(x) = \frac{1}{x}$ to calculate the volume of a sphere. If $g(x) = x + 2$, what is $\left(\frac{f}{g}\right)(x)$?
(A) $\frac{1}{x(x+2)}$
(B) $\frac{1}{x-2}$
(C) $\frac{x+2}{x}$
(D) $\frac{x}{x+2}$
(E) $\frac{2}{x}$

Q6. The function $f(x) = x^2 - 4$ represents the cost of building a road. If $g(x) = x + 1$, what is $(f + g)(x)$?
(A) $x^2 + x - 3$
(B) $x^2 + x - 2$
(C) $x^2 - x - 3$
(D) $x^2 - x - 5$
(E) $x^2 + 2x + 1$

Q7. The ancient Chinese used the function $f(x) = 3x - 2$ to calculate the area of a rectangle. If $g(x) = 2x + 1$, what is $(f - g)(x)$?
(A) $x - 3$
(B) $x + 3$
(C) $x + 1$
(D) $x - 1$
(E) $2x + 1$

Q8. The function $f(x) = x^2$ represents the number of stones needed to build a wall. If $g(x) = x - 2$, what is $(f \cdot g)(x)$?
(A) $x^3 - 2x^2$
(B) $x^3 + 2x^2$
(C) $x^2 - 2x$
(D) $x^3 - 2x$
(E) $x^2 + 2x$

Open Ended Questions (FRQ)

Q9. If $f(x) = x^2 + 2x - 1$ and $g(x) = x - 3$, find $(f \cdot g)(x)$ and determine its domain.

Q10. The ancient Romans used the function $f(x) = \frac{1}{x+1}$ to calculate the volume of a cylinder. If $g(x) = x + 2$, find $\left(\frac{f}{g}\right)(x)$ and simplify the resulting expression.

Chef's Tips

- Tip 1: Use historical context to engage students
- Tip 2: Ensure students show their work and simplify expressions
- Tip 3: Encourage students to use different methods to check their answers